

Electricity Theft Detection in AMI Using Customers' Consumption Patterns

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Abstract—As one of the key components of the smart grid, advanced metering infrastructure brings many potential advantages such as load management and demand response. However, computerizing the metering system also introduces numerous new vectors for energy theft. In this paper, we present a novel consumption pattern-based energy theft detector, which leverages the predictability property of customers' normal and malicious consumption patterns. Using distribution transformer meters, areas with a high probability of energy theft are short listed, and by monitoring abnormalities in consumption patterns, suspicious customers are identified. Application of appropriate classification and clustering techniques, as well as concurrent use of transformer meters and anomaly detectors, make the algorithm robust against nonmalicious changes in usage pattern, and provide a high and adjustable performance with a low-sampling rate. Therefore, the proposed method does not invade customers' privacy. Extensive experiments on a real dataset of 5000 customers show a high performance for the proposed method.

Index Terms—Advanced metering infrastructure (AMI), energy theft, smart grid.

I. INTRODUCTION

ENERGY theft has been a major concern in traditional power systems worldwide. In the United States (U.S.) alone energy theft was reported to cost the utility companies around \$6 B/year [1]. Implementation of advanced metering infrastructure (AMI) as one of the key technologies in smart grids promises to mitigate the risk of energy theft through its monitoring capabilities and the fine-grained usage measurements. However, application of digital smart meters and addition of a cyber layer to the metering system introduce numerous new vectors for energy theft. While traditional mechanical meters can only be compromised through physical tampering, in AMI the metering data can be tampered with, both locally and remotely, before being sent to the smart meters, inside the smart meters, and over the communication links. Penetration tests have already revealed several vulnerabilities in smart meters [2], [3]. In 2009, an organized energy theft attempt against AMI was

reported by U.S. Federal Bureau of Investigation, which potentially could cost a utility company up to \$400 M annually [4]. Therefore, an energy theft detection system (ETDS) that can efficiently detect energy theft attacks against AMI is urgently needed.

Current AMI ETDSs are mainly categorized into three groups [5]: 1) state; 2) game theory; and 3) classification-based. State-based detection schemes [6]–[9] employ specific devices, like wireless sensors and radio-frequency identification tags, to provide a high-detection accuracy, which however, come with the price of extra investment required for the monitoring system including device cost, system implementation cost, software cost and operating/training cost. In game theory-based methods [10]–[11], the problem of electricity theft detection is formulated as a game between the electricity thief and the electric utility. These methods may present a low cost and reasonable, though not optimal, solution for reducing energy theft. Yet, how to formulate the utility function of all players, including thieves, regulators and distributors, as well as potential strategies is still a challenging issue. Classification-based approaches [12]–[18] take advantage of the detailed energy consumption measurements collected from the AMI. Under normal condition customers' consumption follow certain statistical pattern; irregularities in usage pattern can be a sign of some malicious activities. Data mining and machine learning technologies are used to train a classifier based on a sample database, which is then utilized to find abnormal patterns. Since these techniques take advantage of the readily available smart meter data, their costs are moderate. However, several shortcomings in existing classification-based schemes limit their detection rate (DR) and cause a high false positive rate (FPR).

One problem with utilization of machine learning classifiers in ETDS is data imbalance; i.e., the numbers of normal and abnormal samples are not in the same range. Benign samples are easily available using historic data. Theft samples, on the other hand, rarely or do not exist for a given customer. Besides, due to zero-day attacks in many cases examples of attack class cannot be obtained from the historic data. Lack of a thorough dataset of attack samples limits the DR. Furthermore, classification-based methods are vulnerable to contamination attacks, where by granular changes in data and polluting the dataset an adversary deceives the learning machine to accept a malicious pattern as a normal one.

Another challenging issue that affects the performance of classification-based methods is the fact that several

Manuscript received August 12, 2014; revised February 4, 2015; accepted April 6, 2015. Date of publication May 14, 2015; date of current version December 19, 2015. This work was supported by the Natural Sciences and Engineering Research Council of Canada under Grant STPGP 396838. Paper no. TSG-00811-2014.

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Digital Object Identifier 10.1109/TSG.2015.2425222

nonmalicious factors can alter the consumption pattern. For instance, change of residents, change of appliances, seasonality, etc. If such factors are not dealt with properly, they will result in high FPR. As argued in [19], due to the base-rate fallacy phenomenon, the false alarm rate is the limiting factor for the performance of an intrusion detection system (IDS). This is particularly true for ETDSs where false positives are very costly. Once an energy theft attack is detected, on-site inspection is required for final verification, which is an expensive procedure.

Finally, most existing classification-based methods require a high-sampling rate to achieve an acceptable accuracy. However, an effective ETDS must not jeopardize customers' privacy. As discussed in [20], while there are several techniques to mitigate privacy concerns such as power mixing and data aggregation, the privacy-by-design principle of data minimization is the most effective one. The required sampling rate of smart meters, which enables achievement of smart grid goals like load management and demand response, is still a controversial issue. The higher the sampling rate, the more is the risk of revealing customers' private information. Thus, ETDSs that do not rely on a high-data collection frequency are preferable.

In this paper, we present a consumption pattern-based energy theft detector (CPBETD) that employs novel techniques to overcome the problems associated with existing classification-based ETDSs. In CPBETD, the total consumption of each neighborhood is measured by transformer meters, and is compared with the total amount of usage reported by the smart meters. If at this level a nontechnical loss (NTL) is detected, customers in the area with abnormal patterns will be selected as suspicious users. For each customer, a multiclass support vector machine (SVM) is trained using historic data of the user as well as a synthetic attack dataset. The classifier is then used to decide whether a new sample is normal or malicious. The main contributions to this paper are as follows.

- 1) We design a novel algorithm for detecting energy theft attacks against AMI. CPBETD employs transformer meters along with monitoring of abnormalities in customers' consumption patterns to provide a cost-effective and high-performance solution for energy theft detection. Through application of appropriate clustering techniques and transformer meters, unlike existing classification-based methods, CPBETD is robust against contamination attacks and nonmalicious changes in consumption patterns, and therefore, achieves a higher DR and a lower FPR.
- 2) We address the problem of imbalanced data and zero-day attacks by generating a synthetic attack dataset, benefiting from the fact that theft patterns are predictable. Through extensive experiments we show that this significantly improves the DR and enables the detection of a wide range of attack types.
- 3) We study the effect of sampling rate on detection performance, and show that compared to existing methods, CPBETD provides a higher performance with a lower sampling rate. Hence, it has a smaller effect on customers' privacy.

- 4) We argue that the metrics commonly used for ETDS performance evaluation in the literature are not adequate. While DR and in some cases FPR are used for performance evaluation, we discuss that Bayesian detection rate (BDR) and associated costs of the ETDS should also be considered to justify its application.
- 5) We test the performance of CPBETD with real data from over 5000 customers. The dataset is publicly available and can be used as a benchmark for comparison among different energy theft detection methods. The best and most recent energy theft detection solutions existing in the literature are also tested for comparisons. The results prove the effectiveness of our approach.

The rest of this paper is organized as follows. In Section II, we survey the literature related to electricity theft detection in AMI. Threat models for electricity theft attacks are provided in Section III. Section IV describes the CPBETD algorithm. A theoretical analysis of our approach is provided in Section V. In Section VI, we evaluate the performance of the algorithm. Section VII concludes this paper.

II. RELATED WORK

In this section, we review existing ETDSs in the literature, which employ consumption data of smart meters to find fraudulent customers. Monitoring of customers' load profiles for signs of energy theft in traditional power systems had attracted the attention of researchers in the past. In [21], using historical consumption data, a data mining method along with SVM classifier were used to detect abnormal behaviors. The average daily consumptions of customers over a two-year period were calculated and the long term trend in energy consumption was used to detect fraudulent customers. This method is only capable of detecting abrupt changes in load profile. Besides the detection delay is about two years. In [12], using six months usage reports, five attributes including average consumption, maximum consumption, standard deviation, sum of the inspection remarks, and the average consumption of the neighborhood were chosen to create a general pattern of power consumption for each customer. *k*-means based fuzzy clustering was performed to group customers with similar profiles. A fuzzy classification was then performed and Euclidean distances to the cluster centers were measured. Customers with large distances to the cluster centers were considered potential fraudsters. Clustering the customers and relying on long-term measurements limits the accuracy of this ETDS and causes long detection delay. Having more detailed metering information in AMI, CPBETD provides a much better performance with a much shorter delay.

Depuru *et al.* [13]–[15] have studied the use of AMI data in classification-based ETDSs. In [13], SVM was used to approximate the consumption pattern of customers based on 96 readings of smart meters in AMI for each day. The classifier was trained using historical data of normal and theft samples. New samples were classified based on some rules as well as the SVM results. In [14], a neural network model was incorporated to estimate SVM parameters in order to reduce the training time of the classifier, and a data encoding method

was proposed to improve the efficiency and speed of the classifier. Their method, however, is only effective in detecting energy theft attacks that result in zero usage reports, since in one step of the encoding procedure the metering data is converted into binary values. Therefore, the proposed classification technique cannot detect a wide range of attack types. In [15], an improved encoding technique was proposed and both SVM and a rule engine-based algorithm were applied on encoded data to improve the classification accuracy. Portions of the algorithms were parallelized to reduce the detection time. Still the algorithm suffers from the common shortcomings of classification-based methods as we described in detail in Section I. The database used for performance evaluation is not publicly available and the authors did not explain how they obtained theft samples for all customers or what percentages of the training and testing data were theft patterns. On the other hand, in this paper we focus on addressing problems associated with classification-based techniques such as imbalanced data, contamination attacks and dynamic property of consumption patterns to improve the detection accuracy. We also argue that detection time might not be as crucial as other factors like FPR or implementation cost of the ETDS.

In [16], the problem of class imbalance for energy theft detection in traditional power systems was addressed. Considering that the number of people who commit fraud is much less than the number of honest customers, standard classifiers are overwhelmed by benign samples and tend to ignore the minority class. To achieve a higher performance, different classification techniques, including one-class SVM, optimum path forest and C4.5 decision tree, are combined in [16]. The combined classifier showed 2%–10% improvements over individual classifiers. The shortcoming of this method is that while it imposes a high-computational load, the performance improvement is not substantial. We address the problem of class imbalance by generating a dataset of malicious samples using the benign ones. As our experiments show, this method significantly improves the classification performance.

In [7], a multisensor energy theft detection framework for AMI (AMIDS) was presented. AMIDS collects evidences of malicious behavior from three types of information sources: 1) cyber-side network and host-based IDSs; 2) on-meter anti-tampering sensors; and 3) power measurement-based anomalous consumption detectors through nonintrusive load monitoring (NILM). These types of information were combined to minimize the FPR. While combining the information from different sources is effective in reducing the FPR, the algorithms chosen for detecting anomalies have some drawbacks. Use of NILM, which requires a high-sampling rate, reveals information about types and time of use of appliances in customers' premises. In this paper, we focus on detecting energy theft attempts only based on customers' consumption patterns that can also be used as a part of a multisensor framework like AMIDS. Compared to the NILM-based technique CPBETD provides a high performance with a much lower sampling rate.

Salinas *et al.* [17] introduced a privacy preserving ETDS that uses peer-to-peer (P2P) computing. A central meter is deployed in each neighborhood to measure the total electricity

consumption at each time instance, which is assumed to be a linear combination of the energy consumption of all customers in the area. By solving a linear system of equations suspicious users are found. While the proposed method is valuable in that it is the only privacy preserving scheme so far, there are several limitations in this approach. Although the method is vulnerable to inaccuracies in technical loss (TL) calculation, this effect has not been studied. Besides, this method is only effective for detecting energy theft attacks with constant reduction rates, where the real consumption values are multiplied by a constant less than one for several consecutive readings. However, there are many other energy theft scenarios, such as sending random numbers as data. Our approach is capable of detecting more diverse attack types. CPBETD also employs central meters but in a completely different approach. Here, we use distribution transformer meters to short list areas with high probability of theft, and to overcome the limitations of classification-based techniques.

Mashima and Cárdenas [18] suggested modeling the probability distributions of the normal and malicious consumption patterns, and application of the generalized likelihood ratio (GLR) test to detect energy theft attacks. They used autoregressive moving average (ARMA) to model customers' normal and malicious consumption distributions. They assumed that an attacker would choose a probability distribution that decreases the mean value of the real consumption. This, however, is not necessarily true with AMI. Considering the dynamic pricing in smart grids, by only changing the order of meter readings without altering the average, electricity theft is possible. Another major issue with ARMA–GLR detector is that it is only effective if the normal electricity theft behavior and attack patterns can accurately be modeled by an ARMA process. Mashima and Cárdenas [18] also studied nonparametric algorithms such as exponentially weighted moving average, which provide a better precision when the model assumptions are inaccurate. They observed a better performance for ARMA compared to other models. Our tests show that CPBETD significantly outperforms the ARMA–GLR detector.

In our previous work [22], we introduced algorithms for detecting fraudulent activities against AMI, based on finding anomalies in consumption patterns. The present work is an extension of [22]. While [22] covered different types of malicious activities, this paper is focused on energy theft and is tailored for its unique characteristics. Improvements relative to [22] include employment of mechanisms for making the algorithm robust against nonmalicious changes, application of clustering techniques to enhance the classification accuracy, studying the effect of sampling rate on performance, employment of a real dataset of smart meter readings for performance evaluations, and comparison with other existing ETDSs.

III. THREAT MODEL

The main objective of an energy theft attack is to pay less than the real value for the consumed energy. Energy theft attacks against AMI are categorized into three groups [7].

- 1) *Physical Attacks*: Where illegal customers physically tamper with their meters to report a lower usage,

for instance through application of a strong magnet to cause interference, or by reversing or disconnecting the meters. Customers might also directly wire high-consuming appliances to an external feeder, bypassing the meters.

- 2) *Cyber Attacks*: Can be used within the smart meters or over the communication link to the utility company. Examples include gaining privileged access to the meter firmware, tampering with the meter storage, interrupting measurements, and intercepting the communication link.
- 3) *Data Attacks*: Target the metering values and are enabled through physical and cyber attacks.

Detailed explanation of attacks in each category is available in [7]. CPBETD is designed to detect abnormalities in the consumption patterns. Considering that the outcome of each attack is manipulation of the meter readings, CPBETD is effective in detecting all types of attacks presented above.

IV. CPBETD ALGORITHM

Among the three key elements of information security, i.e., confidentiality, integrity and availability, energy theft attacks target data integrity. The major cost of energy theft is financial loss to the utility company caused by unpaid usage. Therefore, while performance is important, the monetary cost introduced by the detection mechanism must be minimized. For the task of theft detection, false positives can be very expensive, since once a suspected fraudulent customer is detected, on-site inspection is required. On the other hand, factors such as detection delay might not be as critical, since the attacks barely introduce immediate damage and once an attack is detected the associated financial loss can be compensated through appropriate fines. In Section V-A, we argue that the acceptable FPR depends on factors such as theft rate, which diverge over different regions; hence, ETDSs with adjustable FPRs are preferable. CPBETD is designed based on the mentioned properties, and has two phases, i.e., training and application.

A. Training Phase

The training phase of CPBETD algorithm is as follows.

- 1) The algorithm is trained to estimate the TL in transmission lines within the neighborhood area network (NAN), E_{TL} . Several mechanisms exist for this purpose. As a case in point in [23] a method for precise calculation of TL in branches of distribution system was proposed. For each branch a specific circuit is assumed. Using the data of consumed energy and currents collected by smart or traditional power meters and by employing least square regression, the likely resistances at the lines connecting the consumption points to the distribution transformers as well as the nonohmic losses are calculated. These parameters are then used to predict TLs in future time intervals. By comparing the total power loss with the estimated TL, NTL in distribution transformer level is calculated. This method does not require frequent measurements. Besides, it is robust against falsified data caused by energy theft attempts, during the prediction phase. Since the false

data will only result in a lower estimation of TL which increases the gap between the total loss and TL, which indicates the NTL. The only limiting factor in the proposed method is that the data used during the training phase must be genuine, otherwise the model will not be accurate. There are methods [24]–[26] that mostly rely on the physical characteristics of the distribution lines rather than customers data to calculate the TL in each segment. Yet they are less accurate due to the dependency of such features on environmental conditions and inaccurate knowledge of the circuit elements. While in most works a high precision was reported, a thorough quantitative evaluation was missing. However, since the uncertainties in TL are usually less than the alterations in customers consumptions, a higher precision for calculating TL compared to NTL is expected. The average calculation error in [26] was reported to be around 0.5%. Design of the TL estimator is outside the scope of this paper. We assume that DR_{TM} and FPR_{TM} are the DR and FPR of the NTL detector.

- 2) The next step is data preprocessing, including operations like dimension reduction (the process of reducing the number of random variables) and normalization. Each data vector in the dataset includes the meter readings of a customer over a 24-h period, for instance for n measurements per hour the data vector has $24 \times n$ elements. While there are several methods for dimension reduction that minimize the information loss by extracting the important features of data, we sum up the in-between samples to make the algorithm compatible with different metering rates. That is, rather than applying a feature extraction technique that saves the key information of a higher dimension data vector while reducing the data size, we only add up the samples. This is because we assume that the time interval of the data received from smart meters is not known.
- 3) Once the data is converted into the proper format, the k -means algorithm [27] is performed on the benign dataset. Several nonmalicious factors can alter the consumption pattern, such as seasonality, change of appliances, and different usage during weekdays and weekends. In order to have a better DR, k -means clustering with different values of k is performed on the data, and each time the silhouette value of the clusters is calculated. A peak in the silhouette plot for $k = l$ shows that the data is originated from l different distributions. Clusters that have few members are eliminated and will not be used for training the classifier. This can help to prevent pollution of the benign dataset by undetected attacks. We use l' to denote the final number of clusters after eliminating the small groups.
- 4) The next step is preparing a dataset for training the classifier. While a dataset of benign samples for each customer is easily obtainable using historic data, malicious samples might not be available, since energy theft might never or rarely happen for a given customer. In order to address the problem of imbalanced data, one solution is the application of single-class classification

techniques where the classifier is trained only using normal samples. However, as shown in Section VI-A for the present application the performance of one-class classifier is poor. Another solution is to utilize density function approximation methods as in [28]. However, such approaches are only effective if the data can accurately be modeled by the approximation function, which in practice might be difficult to attain. Instead, we propose to create a dataset of malicious samples using the benign samples. The goal of energy theft is to report a lower consumption than the true amount used by consumer, or to shift the high usage to low-tariff periods. Thus, it is possible to generate malicious samples using the benign ones. Assuming that $\mathbf{x} = \{x_1, \dots, x_n\}$ is a vector of true consumption values for a 24-h period with n samples, and $\mathbf{x} \in X$ where X is a random vector with P_0 distribution, the utility will receive $\mathbf{y} = \{y_1, \dots, y_n\}$ as the meter readings. For honest customers $\mathbf{y} = \mathbf{x}$ while for fraudulent users $\mathbf{y} = h(\mathbf{x})$, where $\mathbf{y} \in Y$ and Y is a random vector with P_1 distribution so that $E[Y] \leq E[X]$. Through studying different scenarios for energy theft and their effects on measured values, $h(\cdot)$ is extractable. For instance $h(\mathbf{x}) = \alpha\mathbf{x}$, $0 \leq \alpha < 1$ is one possibility. Therefore, it is practical to generate malicious samples using the benign dataset. Although defining all possible functions that result in $E[Y] \leq E[X]$ might not be practical, through considering a variety of scenarios and taking advantage of generalization property of SVM, a thorough dataset of attack samples can be generated.

- 5) The next step is training the classifier. Among several existing techniques for regression and classification, we chose SVM due to its superior performance in many applications compared to traditional methods like likelihood ratio test and neural networks. Furthermore, successful utilization of SVM for problems that deal with the similar data type [21] motivates us to use this classifier. Here, the classifier has $l'+1$ classes, l' for benign and one for malicious samples.
- 6) SVM parameters can be adjusted to attain different performance in terms of DR/FPR. However, as shown in Section VI, the receiver operating characteristic (ROC) of an SVM by itself does not provide a wide range for DR/FPR. In order to achieve the appropriate performance, first the required FPR, FPR_{req} , is calculated as described in Section V-A. Let DR_{SVM} and FPR_{SVM} denote the best achievable performance by SVM, parameter m is calculated as follows:

$$m = -\frac{\log(\text{FPR}_{\text{req}})}{\text{FPR}_{\text{TM}} \times \text{FPR}_{\text{SVM}}} \quad (1)$$

where m represents the number of times an abnormal pattern must be detected before an energy theft alarm is raised. This mechanism, however, will also reduce the DR

$$\text{DR} = (\text{DR}_{\text{TM}} \times \text{DR}_{\text{SVM}})^m \leq \text{DR}_{\text{TM}} \times \text{DR}_{\text{SVM}}. \quad (2)$$

The larger is the value of m , the smaller is the FPR, yet it also means a larger detection delay.

B. Application Phase

The application phase of CPBETD algorithm is as follows.

- 1) For each neighborhood one or more transformer meters measure the total electricity provided to the customers in the area, $E_{\text{TM}}(t)$. This value is compared with the total amount of consumption reported by the smart meters of the corresponding distribution transformer, $\sum_i E_{\text{SM}_i}(t)$. NTL is reported if for any time, t , during a day, $E_{\text{TM}}(t) > \sum_i E_{\text{SM}_i}(t) + E_{\text{TL}}(t) + \varepsilon$, where ε is the calculation error for TL. The above test is performed each time new samples are collected.
- 2) Each new sample is preprocessed and converted into the proper format consistent with the training set.
- 3) SVM is applied to a new sample to determine whether it belongs to the benign or attack class.
- 4) If step 1 did not detect an anomaly and the new sample was classified as benign by the SVM, the new sample is added to the benign dataset and the corresponding attack patterns will be generated and added to the attack dataset.
- 5) If NTL was detected in step 1 and the classifier recognized an attack, a suspicious behavior of the smart meter is reported. An energy theft is detected when the suspicious behavior of the smart meter is repeated m times over a certain period. During this time new samples are stored in a temporary database. Once an energy theft is detected, an appropriate action such as on-site inspection is exercised. Based on the amount of NTL calculated in step 1, higher priority for inspection is assigned to smart meters located in areas with larger NTL. If a theft is verified, samples in the temporary database are added to the attack dataset. Otherwise, they will be added to the benign dataset and their corresponding attack patterns to the attack dataset.
- 6) Another possibility is when step 1 does not detect an NTL, but SVM recognizes an anomaly. This condition might have three causes. It might be due to SVM misclassification, or because of error in NTL calculation in step 1. These cases are not expected to happen frequently in consecutive days. On the other hand, the condition might be because of some alterations in consumption habits; for instance due to changes of residents or appliances, etc., which result in major changes of usage pattern. In this case, the condition will persist. Therefore, when SVM detects an anomaly while there is no sign of NTL in step 1, the new sample is stored in a temporary database. If in the following days this condition happens frequently, the old dataset will be discarded and a new dataset based on the samples in the temporary database will be generated. Once the dataset is large enough, the classifier is retrained. A credibility factor, cf_i , is assigned to each smart meter, which is a binary variable initialized to one. When a nonmalicious anomaly is detected, as described above, cf_i is set to zero and when the situation is resolved it will be set back to one. When the algorithm detects an energy theft, smart meters with $\text{cf}_i = 1$ have a higher priority for further action. This step of the algorithm makes CPBETD robust against nonmalicious changes in consumption pattern.

- 7) If NTL was detected in step 1, but SVM did not recognize any anomaly and the condition persists, it shows that an attack might be happening, but SVM cannot identify it. In this case, the benign dataset of the customers is analyzed for sign of data contamination attack, in which by gradual changes in data and polluting the dataset an adversary deceives the learning machine to accept a malicious pattern as a normal one. The long-term trend in daily usage of the customer is studied. A descending slope in long-term consumption curve can be a sign of contamination attack. If analysis of historic data does not show a contamination attack, CPBETD raises an alarm to indicate that an attack might be happening but the algorithm cannot detect it, and the algorithm continues its normal operations for new samples. This happens in rare scenarios, e.g., when a new load with high consumption is directly connected to a feeder. This step of the algorithm makes CPBETD robust against contamination attacks.

Pseudocodes of the algorithm for the application phase are provided in Fig. 1.

V. THEORETICAL ANALYSIS

A. Required Performance

As explained in [29], the revenue of a distribution utility is formulated as follows:

$$R(e, D) = \sum_i T(q^i - q_u^i) + \sum_i \rho(e, q_u^i, D) F(q_u^i) \quad (3)$$

where q^i is the total consumption of user i , q_u^i is the unbilled part of the usage, T is the price of electricity, ρ denotes the probability of detecting an electricity theft, e is the effort invested in anti-fraud technologies, D is the anomaly detection test, and $F(\cdot)$ represents the recovered fines from the detected theft. On the other hand, the associated costs include the investments for protecting against theft, $\Psi(e)$, and the costs of producing enough electricity to meet the demand of all customers $C(\cdot)$. Therefore, the total profit of the utility is

$$R(e, D) - C(q^i) - \Psi(e). \quad (4)$$

The operational cost of handling D is proportional to the effort, e , of the utility to manage false alarms. At the same time, increasing e improves the probability of detecting a theft. Therefore, the optimal D is the one with maximum ρ subject to an upper bound on FPR. The required FPR might vary over different regions due to the base-rate fallacy phenomenon. The BDR is defined as

$$P(I|A) = \frac{P(I) \times DR}{P(I) \times DR + P(\bar{I}) \times FPR} \quad (5)$$

where I stands for intrusion and A for alarm. For IDS alarms to be reliable, this probability must be high. As (5) suggests, BDR not only depends on DR and FPR, but also is affected by the probability of occurrence of an intrusion. When this probability is low, no matter how big the DR is, the denominator will be subjugated by the FPR term. Therefore, in order to achieve a reasonable value for BDR, very low FPR is required.

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Input:  $NS$ (new sample),  $threshold_1$ ,  $threshold_2$ ,  $threshold_3(> threshold_2)$ 
Output:  $attack$ (boolean)
Variables:  $counter_1$  (number of times an attack is detected),  $counter_2$  (number of non-malicious anomalies),  $NTL$ (boolean),  $CF_i$ (binary),  $SVM_{out}$ (boolean),  $TDB_1$ (temporary database),  $TDB_2$ (temporary database)
if  $E_{TM} > \sum_i E_{SM_i} + E_{TL} + \varepsilon$ 
     $NTL = \text{true}$ ;
else
     $NTL = \text{false}$ ;
end if;
classify  $NS$  by SVM, if it is classified as attack, then  $SVM_{out} = \text{true}$ , otherwise  $SVM_{out} = \text{false}$ ;
if  $NTL = \text{true}$ 
    if  $SVM_{out} = \text{true}$ 
        if  $counter_1 \geq threshold_1$ 
             $attack = \text{true}$ ;
            add  $TDB_1$  to the attack dataset;
            go to end;
        else
             $counter_1++$ ;
            add  $NS$  to  $TDB_1$ ;
        end if
    else if  $SVM_{out} = \text{false}$ 
        Raise an alarm informing about the risk of an undetectable attack;
    end if
else if  $NTL = \text{false}$ 
    if  $SVM_{out} = \text{true}$ 
        if  $counter_2 < threshold_2$ 
             $counter_2++$ ;
            add  $NS$  to  $TDB_2$ ;
        else if  $threshold_2 \leq counter_2 \leq threshold_3$ 
             $counter_2++$ ;
             $CF_i = 0$ ;
            add  $NS$  to  $TDB_2$ ;
        else if  $counter_2 > threshold_3$ 
            discard the old dataset;
            generate a new dataset using  $TDB_2$ ;
            retrain the SVM;
             $CF_i = 1$ ;
        end if;
    else if  $SVM_{out} = \text{false}$ 
         $attack = \text{false}$ ;
    end if;
end

```

Fig. 1. Pseudocodes of application phase.

In the case of energy theft $P(I)$ is usually a small value and might significantly vary in different areas.

B. Classification Method

In SVM, a model is generated based on training data and used to predict the target values of the test data. First, using a kernel function, the data vector is mapped to a higher dimensional space where data classes are more distinguishable. Then a separating hyper-plane with maximum margin to the closest data points in each class is found. This concept translates into a convex quadratic optimization problem formulated in

$$\begin{aligned}
 &\min \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i e_i \\
 &\text{such that } y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - e_i \\
 &\phi(\mathbf{x}_i) = e^{-\gamma \|\mathbf{x}_i\|^2} \quad (6)
 \end{aligned}$$

where $\mathbf{x}_i$ is the i th input data, y_i is the class label for $\mathbf{x}_i$, $\mathbf{w}$ is the normal vector to the hyper-plane, C is the penalty for the error term, e_i measures the constraint violation, ϕ is the kernel function, and b determines the offset of the hyper-plane from the origin along the normal vector [30].

C. Clustering Method

Silhouette plots are applied to determine the number of clusters within a dataset. Assume that the data have been clustered into k clusters and for each sample i , $a(i)$ is the average dissimilarity of i with other samples within the same cluster. Also $b(i)$ is the least average dissimilarity of i to any other clusters. The silhouette value, $s(i)$ is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}. \quad (7)$$

The average of $s(i)$ over all samples within a cluster shows how close the samples in the cluster are, and averaging over the entire dataset shows how properly the data have been clustered. We use this method to determine the number of clusters in the dataset of each customer. Defining separate classes for distinct clusters can help to achieve a higher classification accuracy.

VI. EVALUATIONS

We used the smart energy data from the Irish Smart Energy Trial [31] in our tests. The dataset was released by Electric Ireland and Sustainable Energy Authority of Ireland (SEAI) in January 2012. It includes half hourly electricity usage reports of over 5000 Irish homes and businesses during 2009 and 2010. Customers who participated in the trial had a smart meter installed in their homes and agreed to take part in the research. Therefore, it is a reasonable assumption that all samples belong to honest users. The large number and variety of customers, long period of measurements and availability to the public make this dataset an excellent source for research in the area of analysis of smart meters data. For each customer there is a file containing half hourly metering reports for a 535 day period. We reduced the sampling rate to one per hour and for each customer divided the file into a dataset of 535 vectors, each with 24 components. Then based on the dataset of benign samples, for each sample $\mathbf{x} = \{x_1, \dots, x_{24}\}$, we generated six types of malicious samples as follows for $t = 1, \dots, 24$:

- 1) $h_1(x_t) = \alpha x_t$, $\alpha = \text{random}(0.1, 0.8)$;
- 2) $h_2(x_t) = \beta_t x_t$

$$\beta_t = \begin{cases} 0 & \text{start-time} < t < \text{end-time} \\ 1 & \text{else} \end{cases}$$

start-time = random(0, 23-minOffTime)
duration = random(minOffTime, 24)
end-time = start-time + duration
here minOffTime = 4;

- 3) $h_3(x_t) = \gamma_t x_t$, $\gamma_t = \text{random}(0.1, 0.8)$;
- 4) $h_4(x_t) = \gamma_t \text{mean}(\mathbf{x})$, $\gamma_t = \text{random}(0.1, 0.8)$;
- 5) $h_5(x_t) = \text{mean}(\mathbf{x})$;
- 6) $h_6(x_t) = x_{24-t}$.

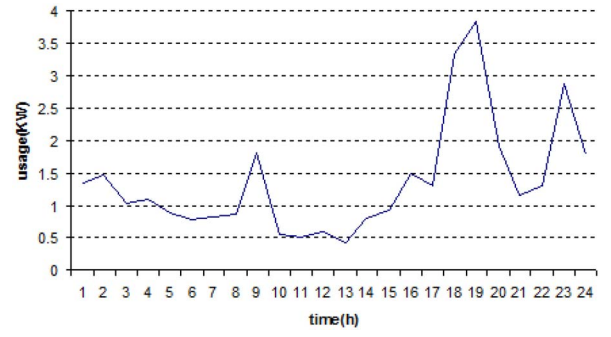


Fig. 2. Example of the daily consumption.

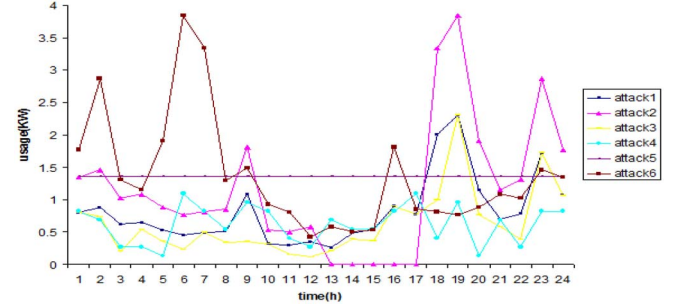


Fig. 3. Example of attack patterns.

In the above, $h_1(\cdot)$ multiplies all the samples by the same randomly chosen value. Using $h_2(\cdot)$ the smart meter does not send its measurements or sends zero for a random duration. $h_3(\cdot)$ multiplies each meter reading by a different random number. $h_4(\cdot)$ and $h_5(\cdot)$ orderly report a factor and the exact value of the average of readings over the day. $h_6(\cdot)$ reverses the order of readings. $h_5(\cdot)$ and $h_6(\cdot)$ represent attacks against load control mechanisms in which the price of electricity varies over different hours of the day; while the total amount of electricity usage stays the same, the usage is reported to happen during the low-tariff periods. An example of the daily consumption of a customer is shown in Fig. 2. Fig. 3 shows the corresponding attack patterns. Experiment results in the rest of this section are for the anomaly detector part of CPBETD. The total DR and FPR of the algorithm are scaled by the DR_{TM} and FPR_{TM} .

A. Experiment 1: One-Class SVM

In the first test we examine the performance of one-class SVM, in which the classifier is trained only using normal samples. Among 535 samples of the benign dataset we use 458 samples for training and 77 samples for testing. In each seven consecutive days one sample is randomly chosen for the testing set and the other six for the training set. Also for each attack type, 535 samples are generated. In total, the testing set includes 3287 samples, scaled in the range $[-1, 1]$. We choose radial basis function (RBF) as the SVM kernel, and use grid search [30] to find the best values for the kernel parameter, γ , and the upper bound in the fraction of training points, ν . Other SVM parameters are $C = 50$ and $e = 0.1$. The above procedure is repeated for 5000 customers. For each

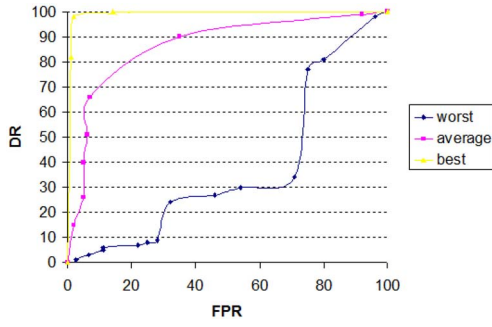


Fig. 4. ROC curves for three customers with the worst, average, and best detection performance using one-class SVM.

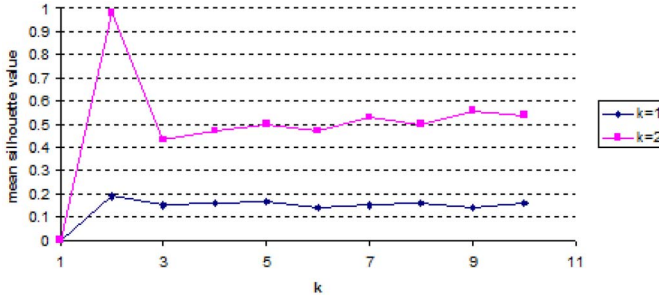


Fig. 5. Silhouette plots for customers with $k = 1$ and $k = 2$.

customer the best performance is considered the one that provides the highest difference (HD) between DR and FPR for different combination of γ and ν . The average value of HD for 5000 customers is found to be 47%, with the average DR and FPR being 76% and 29%, respectively. Fig. 4 shows the ROC curve for three customers with best, average and worst performances. As the results suggest without the abnormal samples in training phase, the classification performance is not promising.

B. Experiment 2: Multiclass SVM

In the second experiment, we train the classifier using both benign and malicious samples. After preprocessing we employ k -means clustering on the benign dataset and study the silhouette plots to find the best value for k . For most customers we observe that $k = 1$ or 2 provides the best result. Then we train a multiclass SVM with $k + 1$ classes. The silhouette plots for two customers with $k = 1$ and $k = 2$ are shown in Fig. 5. Originally, the training set includes 458 benign and 2748 malicious samples. We use over sampling, in which the members of the minority class are replicated, to make the number of benign and attack samples equal. This helps to avoid the classifier from being biased toward the larger class. We use RBF kernel and apply grid search to find the appropriate values for γ and C ; the value of ϵ is 0.1. For this experiment we observe the average HD of 83% with the average DR and FPR being 94% and 11%, respectively. Fig. 6 shows the ROC curve of three customers with best, intermediate, and worst performances. As we can see, multiclass SVM provides a significantly better performance compared to the single class SVM.

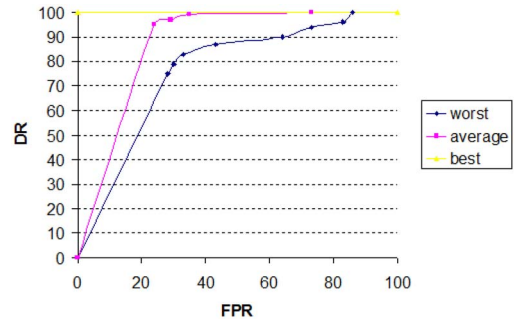


Fig. 6. ROC curves for three customers with the worst, average, and best detection performance using multiclass SVM.

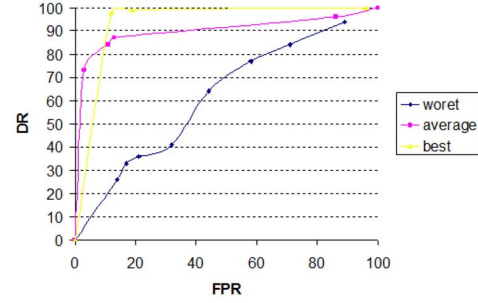


Fig. 7. ROC curves for three customers with the worst, average, and best detection performance using multiclass SVM with undefined attacks.

TABLE I
EXPERIMENT RESULTS

Performance	Exp. 1	Exp. 2	Exp. 3
HD(%)	47	83	70
DR(%)	76	94	86
FPR(%)	29	11	16

C. Experiment 3: Multiclass SVM With New Attacks

The attacker might use a different $h(\cdot)$ than those used to train the classifier. To study this effect, in this experiment we only train the SVM using $h_3(\cdot)$, yet we include all $h_i(\cdot)$ s in the test set. The SVM parameters are similar to Experiment 2. We observe the average performance of HD = 70% with DR = 86% and FPR = 16%. Three examples of ROC curves are shown in Fig. 7. Though in this experiment we observe a lower DR compared to Experiment 2, the performance is still significantly better than one class SVM. A summary of the experiment results is provided in Table I.

D. Overall Performance of CPBETD Algorithm

While the exact value for the probability of an energy theft is unknown and varies across different areas, we use the result of an experiment in [32] to evaluate the performance of CPBETD. In 2001, the Arizona public service company conducted a study to provide an accurate estimate of energy theft in its coverage area. The goal of the study was to find the extent of meter tampering and the resulting financial loss with 95% accuracy. Among 868 000 customers, 550 were randomly selected including rural (65%) and urban, residential (88%) and industrial users. The results showed the definite meter tampering rate of 0.72%. Considering the tampering rate of 0.72%,

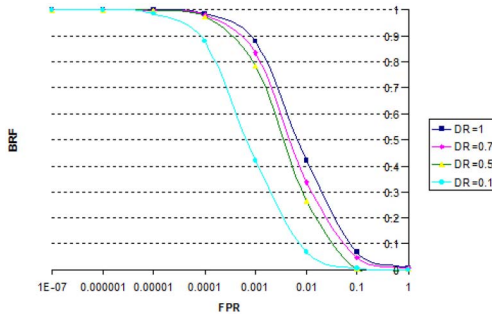


Fig. 8. Relationship between BDR and FPR.

TABLE II
BDR FOR DIFFERENT VALUES OF m

m	1	2	3	4
DR(%)	94	88	83	78
FPR(%)	11	1	0.1	0.01
BDR(%)	6.3	38	85	98

Fig. 8 shows the relationship between BDR and FPR. As the figure suggests, in order to achieve a 90% BDR even for 100% DR, FPR must be around 0.1%. Table II shows the DR, FPR, and BDR for different values of m . By increasing m , appropriate values for FPR and therefore BDR are achievable, while DR stays in an acceptable range. Considering the ROC curves of the SVM, obtaining such a low FPR for $m = 1$ is either impossible or happens for a very low DR.

E. Effect of Sampling Rate on Performance

To study the effect of sampling rate on detection performance, we tested the average performance of 100 customers for different metering frequencies. The result is shown in Table III. We observe a peak at 12 samples/day, which indicates that higher sampling rates do not necessarily result in a better performance. This can be due to the fact that for lower sampling rates, the effect of theft on each sample might be more apparent. For instance, when a fraudulent customer aims to report 5 kW less usage than his actual consumption by uniformly dividing this amount over all samples, for 24 samples the changes in usage pattern is less apparent than when 5 kW is spread over 12 samples. Therefore, the attack might be more apparent for lower sampling rates. On the other hand, for very low-sampling rates the consumption pattern cannot be modeled accurately. Using the lowest sampling rate that provides an acceptable performance is important, not only to minimize the required resources, but also to preserve customers privacy. Greveler *et al.* [33] showed that for 0.5 samples/s, one can find as much detailed information as the title of the movie or the TV channel displayed in the household. McLaughlin *et al.* [7], using 0.5 samples/min, were able to find the type and time of usage of appliances inside the household. This information cannot be obtained precisely using 1 sample/h. For 1 sample/day, one cannot find detailed information about consumption habits of a user, still it is possible to conclude whether or not the customers are at home. This information can be used for example for the

TABLE III
EFFECT OF SAMPLING RATE ON DETECTION PERFORMANCE

Sampling Rate(samples/day)	DR(%)	FPR(%)
1	62.1	28.0
2	76.5	16.0
4	88.7	9.0
6	90.2	9.7
12	93.1	11.0
24	89.8	12.3

purpose of committing burglary. On the other hand, the utility requires metering data for demand response management. In [11], a game theory approach is proposed to formulate the privacy-preserving demand-response problem as the task of finding the maximum allowable sampling rate that keeps the demand lower than a predefined maximum value. We cannot say there is a specific point at which the users' measurement is or is not private anymore. However, the lower the sampling rate the less information can be extracted from the measured data. So the sampling rate that allows the utility to achieve its goals regarding demand response, etc. can be found and the utilities can be forced by law to not collect metering data with higher than the required frequency. One advantage of CPBETD is that it can provide a good performance even for the sampling rate as low as 4 samples/day.

F. Discussion and Comparison

Customers' normal consumption pattern may vary due to several nonmalicious factors. These changes can be temporary, periodic, or permanent and potentially can cause false positives. CPBETD uses mechanisms to handle these situations. Short-term changes are the result of some unusual behaviors, for instance having a big party, which do not last for more than one or few days. Application of distribution transformer meters help to reduce false alarms due to short-term changes since a suspicious behavior is reported only when both distribution transformer meters and SVM detect an anomaly. Therefore, if a customer has a benign unusual behavior while NTL is not detected in distribution transformer level, false positive will not be generated; unless another customer within the neighborhood area is committing fraud at the same time which results in detection of NTL in the distribution transformer. In this case, the customer is considered suspicious. To handle this situation we suggested calculation of the required FPR and adjustment of parameter m so that theft will only be reported when a suspicious behavior is repeated several times. Moreover, customers might have dissimilar usage habits over different days of week (weekday-weekend) or seasons. Using k -means clustering enables the algorithm to separate distinct distributions in data, and separate classifiers can be trained accordingly. If a time dependency is observed for clusters, the corresponding classifiers are labeled, and based on the time of new samples they are classified using the appropriate classifier. Finally, permanent changes can happen due to change of residents, appliances, etc. As explained in step 6 of the application phase, CPBETD can identify these changes and avoid generating false positives.

TABLE IV
COMPARISON AMONG ENERGY THEFT DETECTION METHODS

Parameter	ARMA-GLR[18]	CPBETD	AMIDS [7]	P2P [17]
DR(%)	67	94	NA	attack1: 96 others: NA
FPR(%)	28	11	NA	attack1: 9 others: NA
Required sampling rate	Low	low	High	low
Robustness against non-malicious changes	vulnerable	robust	vulnerable	robust
Privacy preservation	medium	medium	week	strong

It is also possible that the dataset that is initially used to train the algorithm contains theft samples. One solution is harder supervision during the data collection period. Using the NAN NTL detector, areas suspicious to energy theft can be short listed and customers in the area can be considered for on-site inspection. During this phase application of TL calculation methods that do not rely on customers data is recommended.

One limitation of machine learning methods for anomaly detection in the security domain is vulnerability to contamination attacks. Concurrent use of NAN NTL detector along with the customers' anomaly detector in CPBETD helps to mitigate this concern. When an NTL is detected in NAN, while there is no sign of anomaly in customers' usage, the trend in daily usage of customers can be monitored. A descending slope in long-term consumption curve can be an indication of a contamination attack. CPBETD is capable of detecting a variety of attack types. The only energy theft scenario that cannot be detected is when a customer employs a new appliance and directly connects it to an outside feeder. Since there is no sign of this appliance in the historic dataset, the anomaly will not be detected.

Table IV shows a comparison among the proposed algorithm and the most recent and best ETDSs existing in the literature. We implemented the ARMA-GLR detector proposed in [18], and tested its performance using the SEAI dataset. The results are provided in Table IV. The NILM-based technique in AMIDS [7] requires around 0.5 samples/min to function properly, and thus the 2 samples/h sampling rate in the SEAI dataset is inadequate to measure its performance. We also implemented the lower upper decomposition algorithm in [17], which is based on P2P computing. While the algorithm is effective in detecting type 1 attacks, it performed poorly for other attack types.

VII. CONCLUSION

In this paper, we have introduced CPBETD, a new algorithm for detecting energy theft in AMI. CPBETD relies on the predictability of customers' normal and malicious usage patterns. Along with application of SVM anomaly detector, the algorithm uses silhouette plots to identify the different distributions in the dataset, and relies on distribution transformer meters to detect NTL at the transformer level. We have shown that these features provide a high performance and make the

algorithm robust against nonmalicious changes in consumption pattern as well as data contamination attacks. In practice, the required performance for an ETDS may vary across different regions. We have shown that by introducing some delay to the detection algorithm, an adjustable performance to match different objectives is achievable. Using extensive experiments on a real dataset of 5000 customers, we have shown that the proposed algorithm provides a high-performance even with a low-sampling rate, which helps to preserve customers' privacy.

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